

LLM Skill IQ Advanced Practice Pack

Medium, hard, and very hard MCQs for Pluralsight-style LLM assessment preparation

This pack contains 45 practice questions with detailed answer explanations. It is designed to help you score strongly on conceptual and practical LLM assessment areas: architecture, Transformers, RAG, prompting, fine-tuning, evaluation, safety, and deployment.

How to use this pack

- Do 25 questions in 15 minutes to simulate the Skill IQ pace.
- Review every wrong answer and write one-line definitions for the missed concept.
- Repeat the very hard architecture and RAG questions until you can explain why the wrong options are wrong.
- This is original practice material, not an official Pluralsight question dump.

Target topics

Area	What you must know
Architecture	Decoder-only vs encoder-only vs encoder-decoder, causal masks, attention, KV cache, position encodings
LLM training	Pre-training, supervised fine-tuning, instruction tuning, RLHF, preference optimization
Prompting	Zero-shot, few-shot, system prompts, temperature, structured outputs, prompt injection
RAG	Chunking, embeddings, vector DBs, retrieval, reranking, grounding, evaluation
Deployment	Latency, cost, quantization, caching, API vs local model, privacy, monitoring

Medium Questions

Aim: confirm core concepts quickly and accurately.

1. Which architecture most closely describes GPT-style text generation?

- A. Encoder-only Transformer with bidirectional attention
- B. Decoder-only Transformer with causal self-attention
- C. CNN-only architecture with convolutional filters
- D. RNN-only architecture with recurrent hidden states

2. What does an LLM context window mainly limit?

- A. The number of GPUs used during training
- B. The maximum number of tokens the model can consider in one request
- C. The number of parameters the model can update during fine-tuning
- D. The number of users who can call the model at once

3. Why does tokenization matter when using LLM APIs?

- A. It determines the model's database schema
- B. It affects cost, context usage, and how text is split internally
- C. It prevents hallucination completely
- D. It replaces the need for embeddings

4. What is the main role of self-attention in a Transformer?

- A. To compress the model after training
- B. To let each token weigh relevant relationships with other tokens
- C. To store documents in a vector database
- D. To remove the need for positional information

5. Which statement best distinguishes BERT-style and GPT-style models?

- A. BERT is generally encoder-only and GPT is generally decoder-only
- B. BERT is always larger than GPT
- C. GPT cannot process prompts; it can only output random text
- D. BERT and GPT both use CNNs as their main architecture

6. What is the primary goal of RLHF in LLM development?

- A. Increase the context window length
- B. Align model outputs with human preferences and safety expectations
- C. Convert text into image embeddings
- D. Remove the need for supervised fine-tuning

7. What is few-shot prompting?

- A. Training the model on a few examples and saving new weights
- B. Providing examples in the prompt before asking the model to perform a task
- C. Using the smallest available model
- D. Asking the model to answer with fewer words

8. In a basic RAG system, what does the embedding model do?

- A. Converts text chunks and queries into vectors for similarity search
- B. Replaces the LLM generator
- C. Encrypts documents before upload
- D. Guarantees that all answers are factually correct

9. What is an LLM hallucination?

- A. A response that is confidently stated but false or unsupported
- B. A response that is too short
- C. A model using a vector database
- D. A model refusing to answer unsafe content

10. For a deterministic classification or extraction task, which temperature setting is usually preferred?

- A. Very high temperature to maximize randomness
- B. Low temperature to reduce randomness
- C. Temperature has no effect on output variation
- D. Temperature must always be exactly 1.0

11. When is RAG usually a better first choice than fine-tuning?

- A. When the system needs current or document-specific knowledge with traceable evidence
- B. When there are no documents to retrieve from
- C. When the goal is to permanently change every model weight
- D. When the model must ignore external context

12. What is the role of a system prompt in a chat LLM application?

- A. It is a low-priority user message
- B. It provides high-priority behavior and policy instructions to the assistant
- C. It stores embeddings in the vector database
- D. It is only used for image generation models

13. Which model architecture is classically associated with translation tasks in the original Transformer setup?

- A. Encoder-decoder
- B. CNN-only
- C. RNN-only with no attention
- D. Decision tree

14. What does top-k retrieval usually refer to in RAG?

- A. The top k model parameters by magnitude
- B. The number of highest-scoring chunks returned by the retriever
- C. The number of GPUs used for generation
- D. The maximum number of answer tokens allowed

15. Which is the best description of instruction tuning?

- A. Training a model to better follow natural language instructions
- B. Compressing model weights into fewer bits
- C. Changing the tokenizer after deployment
- D. Increasing document chunk size in RAG

Hard Questions

Aim: handle applied architecture, RAG, alignment, and deployment trade-offs.

16. In a decoder-only Transformer, what does the causal attention mask prevent during training or generation?

- A. A token from attending to future tokens that should not yet be visible
- B. A token from attending to any previous token
- C. The model from using positional information
- D. The model from generating punctuation

17. Why can a decoder-only model perform translation even without a separate encoder?

- A. It secretly uses a CNN encoder at inference time
- B. It treats the instruction and source sentence as previous context, then generates the target text as a continuation
- C. It cannot actually translate; all translations come from an external dictionary
- D. It updates its weights every time a user asks for translation

18. What is the purpose of a KV cache during autoregressive decoding?

- A. To store past key/value attention states so they do not need to be recomputed every token
- B. To store training labels for RLHF
- C. To encrypt all prompts at rest
- D. To rank retrieved documents by BM25 score

19. What problem do positional encodings or rotary positional embeddings address?

- A. They tell the model token order or relative position information
- B. They remove all bias from the model
- C. They make embeddings private by default
- D. They convert text into SQL queries

20. What is the main idea behind LoRA or other parameter-efficient fine-tuning methods?

- A. Update small adapter parameters instead of all base model weights
- B. Replace the Transformer with an RNN
- C. Remove the tokenizer from the pipeline
- D. Use only sparse keyword search instead of embeddings

21. What is a common trade-off when quantizing an LLM?

- A. Lower memory and compute cost, with possible quality loss
- B. Higher memory usage and guaranteed quality improvement
- C. No effect on memory but doubles context length
- D. It converts decoder-only models into encoder-decoder models

22. What is a Mixture-of-Experts model at a high level?

- A. A model that routes tokens through a subset of specialized expert networks
- B. A model that always uses every parameter for every token
- C. A model made only from embeddings with no Transformer layers
- D. A model that cannot be used for generation

23. In a RAG system, what is a prompt injection risk from retrieved documents?

- A. A document chunk may contain instructions like 'ignore previous instructions' that try to manipulate the model
- B. The retriever always deletes the system prompt
- C. The vector database permanently changes model weights
- D. Chunk overlap guarantees malicious text cannot be retrieved

24. Which evaluation dimension asks whether the answer is supported by the provided context?

- A. Faithfulness or groundedness
- B. Latency
- C. Throughput
- D. Tokenization speed

25. Why is chunk overlap often used when indexing documents for RAG?

- A. To reduce the chance that important information is split across chunk boundaries
- B. To guarantee zero hallucination
- C. To make all chunks exactly one sentence long
- D. To prevent embeddings from being generated

26. What is a key difference between dense vector retrieval and sparse keyword retrieval?

- A. Dense retrieval uses learned vector representations of meaning; sparse retrieval relies more on exact or weighted terms
- B. Dense retrieval can only search images, while sparse retrieval can only search audio
- C. Sparse retrieval always requires fine-tuning an LLM
- D. Dense retrieval cannot handle synonyms

27. Why might a cross-encoder reranker be slower but more accurate than first-stage vector retrieval?

- A. It jointly scores the query and candidate document text, which is more expensive but more precise
- B. It never reads the query
- C. It only counts exact keyword matches
- D. It removes the need for a generator model

28. Which statement best compares supervised fine-tuning and RLHF?

- A. Supervised fine-tuning teaches from labeled examples; RLHF optimizes against human preference signals
- B. RLHF is the same as tokenization
- C. Supervised fine-tuning requires no data
- D. RLHF is only used for vector databases

29. What is a major privacy consideration when using a third-party LLM API with enterprise documents?

- A. Sensitive data may leave the organization's environment unless controls are in place
- B. API models cannot process text
- C. Third-party APIs always fine-tune on every prompt by default
- D. Privacy is not relevant for RAG

30. In many multimodal LLM systems, how is non-text input commonly connected to the language model?

- A. A modality encoder maps images/audio/video into representations the language model can use
- B. The language model ignores non-text input completely
- C. Images are always converted to SQL tables first
- D. A vector database replaces all Transformer layers

Very Hard Questions

Aim: handle edge cases and advanced reasoning questions that separate high scores from average scores.

31. Which statement about perplexity is most accurate?

- A. Lower perplexity generally means the model assigned higher likelihood to the text, but it does not guarantee better task performance or alignment
- B. Higher perplexity always means better factuality
- C. Perplexity is the same as RAG faithfulness
- D. Perplexity can only be computed for image models

32. What is a common drawback of beam search for open-ended LLM generation?

- A. It can produce less diverse, generic, or overly high-probability text compared with sampling approaches
- B. It prevents the model from using previous tokens
- C. It cannot be used with decoder-only models
- D. It always improves creativity and factuality

33. What is the main efficiency idea behind grouped-query attention compared with standard multi-head attention?

- A. Multiple query heads share fewer key/value heads, reducing KV cache memory and bandwidth
- B. It removes attention entirely
- C. It changes the model into an encoder-only classifier
- D. It guarantees hallucination-free answers

34. Why can a very long context window still fail to answer a question whose evidence is technically present?

- A. Attention and retrieval behavior may not reliably focus on the most relevant evidence, especially with distractors
- B. Long-context models cannot tokenize text
- C. A longer context always guarantees perfect recall
- D. The causal mask hides all previous tokens from the answer position

35. What is the 'lost in the middle' issue in long-context LLM usage?

- A. Models may use information near the beginning or end of context better than information buried in the middle
- B. Models cannot process the middle token of any sentence
- C. Vector databases delete middle-ranked chunks
- D. It is another name for RLHF

36. A RAG answer is fluent and correct according to general knowledge, but the retrieved context does not support it. Which evaluation failure is most directly shown?

- A. Low faithfulness despite possibly high answer correctness
- B. Low latency
- C. High throughput
- D. Low token count

37. Why is fine-tuning usually not the best way to add frequently changing factual knowledge?

- A. Facts can become stale, are hard to trace, and may not be reliably recalled; retrieval is usually better for changing knowledge
- B. Fine-tuning cannot change model behavior at all
- C. Fine-tuning automatically creates citations
- D. Fine-tuning eliminates all privacy concerns

38. What happens if the embedding model used for indexing differs significantly from the one used for querying?

- A. Similarity scores may become unreliable because vectors may not live in the same semantic space
- B. Retrieval will always improve
- C. The LLM automatically converts old vectors into the new space
- D. It only affects answer style, not retrieval quality

39. Which statement about JSON mode or function calling is most accurate?

- A. It can improve output structure, but it does not by itself guarantee factual correctness
- B. It proves the answer is grounded in the context
- C. It replaces the need for validation
- D. It turns a decoder-only model into a database engine

40. What is speculative decoding mainly used for?

- A. Reducing latency by having a smaller draft model propose tokens that a larger model verifies
- B. Increasing factuality by retrieving more documents
- C. Training reward models from human preferences
- D. Preventing all unsafe outputs

41. What is in-context learning?

- A. The model adapts behavior from examples in the prompt without changing its weights
- B. The model permanently trains on the user's current prompt
- C. The model deletes its tokenizer and learns a new one
- D. The vector database rewrites the model parameters

42. In a decoder-only chat model, why can the answer position attend to the entire user question even with a causal mask?

- A. Because the user question tokens come before the answer tokens in the sequence, so they are previous context
- B. Because causal masks are disabled for chat models
- C. Because the model has a hidden encoder not shown to users
- D. Because attention is only used during training, not inference

43. Why does standard full self-attention become expensive as context length grows?

- A. The attention matrix scales roughly with the square of sequence length
- B. The tokenizer stops working after 100 tokens
- C. Longer context always reduces compute
- D. It requires updating model weights during every request

44. What is the main purpose of Maximal Marginal Relevance in retrieval?

- A. Balance relevance and diversity among selected documents or chunks
- B. Increase the temperature of generation
- C. Replace the language model with a CNN
- D. Encrypt embedding vectors

45. What is catastrophic forgetting in the context of fine-tuning?

- A. A model loses some previously learned capabilities after being trained too narrowly on new data
- B. A model refuses all unsafe prompts
- C. A vector database returns no chunks
- D. A tokenizer splits rare words into subword tokens

Answer Key with Explanations

Use this after timed practice. Do not check explanations while answering.

Medium

1. **B.** GPT-style models are decoder-only Transformers. They generate text autoregressively by predicting the next token from previous tokens.
2. **B.** The context window is the token budget available for prompt, retrieved context, tool results, and output.
3. **B.** APIs usually price and limit by tokens. Tokenization also affects how languages, code, and symbols are represented.
4. **B.** Self-attention builds contextual representations by letting tokens focus on other relevant tokens.
5. **A.** BERT is commonly used as an encoder-only model for understanding tasks. GPT is commonly used as a decoder-only model for generation.
6. **B.** Reinforcement Learning from Human Feedback is used to make responses more helpful, aligned, and preferred by human raters.
7. **B.** Few-shot prompting includes example input-output pairs inside the prompt. It does not update model parameters.
8. **A.** Embeddings represent semantic meaning numerically so relevant chunks can be retrieved with vector search.
9. **A.** Hallucination means the model generates information that sounds plausible but is incorrect or not grounded in evidence.
10. **B.** Lower temperature usually produces more stable, less random outputs, which is useful for extraction and classification.
11. **A.** RAG is strong for changing knowledge, enterprise documents, citations, and evidence-grounded answers.
12. **B.** The system prompt sets the assistant's role, constraints, and behavior before user messages are considered.
13. **A.** The original Transformer used an encoder-decoder design, which is natural for source-to-target tasks like translation.
14. **B.** Top-k retrieval returns the k most relevant chunks according to a retrieval score.
15. **A.** Instruction tuning uses instruction-response examples so the model learns to follow commands more reliably.

Hard

16. **A.** Causal masking enforces next-token prediction by blocking attention to future positions.
17. **B.** Decoder-only LLMs frame many tasks as sequence completion: prompt plus source text in, target text out.
18. **A.** A KV cache improves decoding efficiency by reusing previous attention keys and values.
19. **A.** Self-attention alone is largely position-agnostic, so models need a way to represent token positions.
20. **A.** LoRA freezes the base model and trains small low-rank matrices, reducing compute and storage cost.
21. **A.** Quantization stores weights at lower precision, reducing resource use but sometimes lowering output quality.
22. **A.** MoE models can increase total parameter count while activating only selected experts per token.
23. **A.** Retrieved content is untrusted input. The model must treat it as evidence, not as higher-priority instructions.
24. **A.** Faithfulness checks whether generated claims are justified by retrieved or supplied evidence.
25. **A.** Overlap preserves context around boundaries, improving retrieval when relevant evidence spans adjacent text.
26. **A.** Dense retrieval is semantic, while sparse methods like BM25 are term-based. Hybrid search can combine both.
27. **A.** Rerankers can examine query-document interactions directly, improving ranking quality at higher compute cost.
28. **A.** SFT learns from example outputs. RLHF uses preference/reward feedback to shape model behavior.
29. **A.** Organizations must consider data handling, retention, logging, encryption, and contractual/API controls.
30. **A.** Multimodal systems often use encoders/projectors to translate non-text inputs into representations compatible with the LLM.

Very Hard

31. **A.** Perplexity measures language modeling likelihood, but high-quality instruction following and factuality need separate evaluation.

- 32. A.** Beam search optimizes likely sequences, which can be useful for constrained tasks but may reduce diversity in open-ended generation.
- 33. A.** Grouped-query attention reduces the number of key/value heads stored and read during decoding while keeping multiple query heads.
- 34. A.** Long context increases capacity, but it does not guarantee the model will use the right evidence. Retrieval and placement still matter.
- 35. A.** Relevant evidence placed in the middle of a long prompt can be less likely to influence the answer than evidence at the edges.
- 36. A.** Faithfulness is about support from context. An answer can be correct in the world but still not grounded in the retrieved evidence.
- 37. A.** Fine-tuning can shape behavior, tone, and task format, but changing facts are better handled through retrieval or tools.
- 38. A.** Embeddings from incompatible models are not safely comparable. Re-indexing is often required when changing embedding models.
- 39. A.** Structured output controls format. You still need validation, grounding, and tool-result checks for correctness.
- 40. A.** Speculative decoding is an inference acceleration technique. It aims to speed up generation, not improve reasoning quality directly.
- 41. A.** In-context learning changes behavior through prompt examples at inference time; it does not update model weights.
- 42. A.** Causal masking blocks future tokens. The prompt is earlier in the sequence, so generated answer tokens can attend to it.
- 43. A.** Full attention compares many token pairs, so cost and memory can grow quadratically with sequence length.
- 44. A.** MMR avoids returning many near-duplicate chunks by selecting results that are relevant but also diverse.
- 45. A.** Overly narrow or aggressive fine-tuning can degrade general capabilities learned during pre-training.

Architecture Cheat Sheet

Concept	High-score explanation
Encoder-only	Reads the full input bidirectionally. Strong for classification, embeddings, search, and understanding tasks. Example: BERT-style models.
Decoder-only	Uses causal self-attention and predicts the next token. Strong for chat and open-ended generation. Example: GPT, Llama, Mistral-style models.
Encoder-decoder	Encoder processes input; decoder generates output. Natural for translation and summarization. Example: T5, BART, original Transformer.
Causal mask	Prevents a token from attending to future tokens. The answer can still attend to the user prompt because the prompt is previous context.
KV cache	Stores previous key/value attention states so generation does not recompute the whole prefix every time.
Self-attention	Learns which earlier tokens matter for the current token. This is why decoder-only models can interpret prompts.
Positional encoding	Adds order information because attention by itself does not know token positions.
MoE	Routes tokens through selected expert networks. Total parameters can be high while active parameters per token are lower.

Final high-score checklist

- Explain why GPT-style models are decoder-only yet still understand prompts.
- Explain causal masking, KV cache, context window, and token-by-token generation.
- Know when to choose RAG, fine-tuning, prompt engineering, or tool calling.
- Know RAG failure modes: bad chunking, poor embeddings, missing evidence, prompt injection, and low faithfulness.
- Know deployment trade-offs: latency, cost, quantization, API privacy, monitoring, and caching.

Source notes

- Pluralsight states Skill IQ assessments include about 25 questions and take about 10-15 minutes.
- Pluralsight's Large Language Models path lists AI language models and foundation models, RLHF, NLP and Transformer models, and practical LLM applications as learning areas.
- Hugging Face documentation describes Mistral models as decoder-only Transformer models.
- These practice questions are original and are not official Pluralsight assessment items.